

Linear Algebra – Problem Set 3

Linear Independence, Bases & Dimension · 12 problems · Original problems in the spirit of MIT 18.06 (Strang)

Problem 1

In $\mathbb{R}^3$, decide whether the three vectors are linearly independent, and justify it by elimination rather than by eye:

Problem 2

In $\mathbb{R}^3$ consider

Problem 3

True or false? Give a counterexample or a correction for each false one.

Problem 4

Let $u = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $v = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$, $w = \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix}$ and $z = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$.

Problem 5

In the plane, let $a = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $b = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

Problem 6

Let $W = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$.

Problem 7

In $\mathbb{R}^3$ with respect to the basis $B = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$:

Problem 8

Let B be the basis from Problem 7 and let $C = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$.

Problem 9

Identify every polynomial of degree ≤ 2 with its coefficient vector relative to $(1, x, x^2)$ — the polynomial $a + bx + cx^2$ becomes (a, b, c) . Let

Problem 10

An experiment records a temperature T and a pressure P :

Problem 11

Prove that a linearly independent set in $\mathbb{R}^n$ contains at most n vectors. Give two proofs:

Problem 12

For a square matrix A , prove the chain of equivalences